MA480

Assignment 1

* Solve the following ODEs

① $\frac{dy}{dx} = \sqrt{x+y}$ reducible to separable form

② $(x^2+y^2)\,dy + 2x(x+y)\,dx = 0$ homogeneous

③ $(3x+y-3)\,dy = (2x+y-2)\,dx$ reducible to homogeneous form.

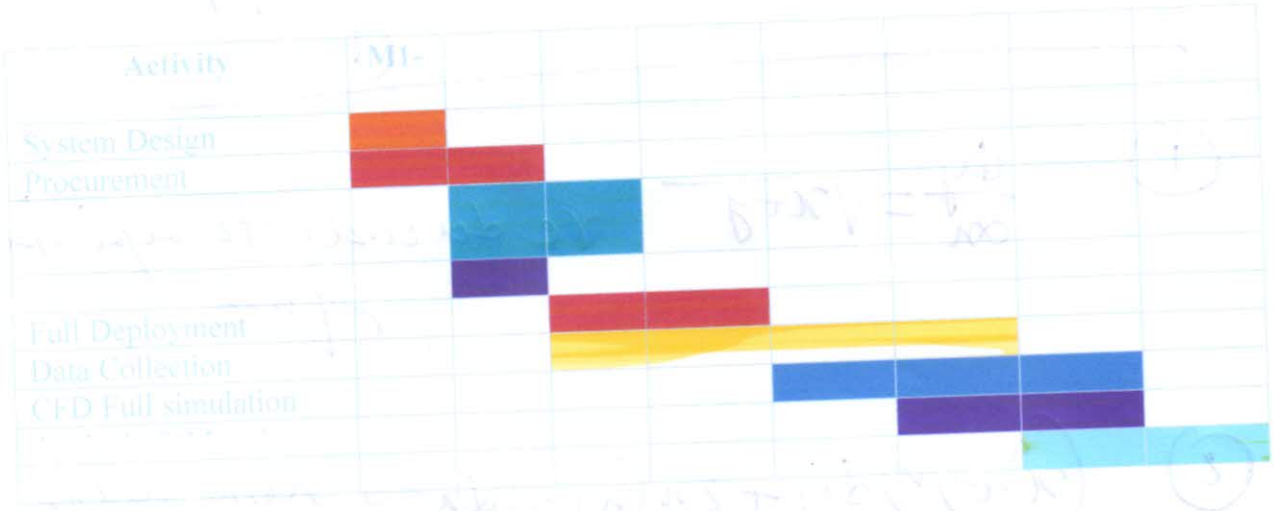
④ $\frac{dy}{dx} - \frac{\tan y}{1+x} = (1+x)e^x \sec y$ non-linear ODE

⑤ $(D^2 - \cancel{4y} \; 4D + 4)y = x^3$ inverse operator method

⑥ $(D^2-1)y = e^x x^2 \sin x$ " " " .

⑦ $y'' - 2y' + 2y = e^x \sec x$ by VPM

⑧ $(x^2D^2 - xD - 3)y = x^2 \log x$



$$\cancel{\beta} = \frac{15 \times 1}{}$$

$$12050 = \frac{15}{v}$$

$$\Rightarrow v = \frac{15}{12050}$$